

High-Rank Adaptation May Be Useful For Recovering Audio Dropout

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Abstract

Automatic speech recognition is inherently vulnerable to a lot of different variations of noise that can corrupt the data. In an ideal world if any of this information was missing, there would be an opportunity to repeat or recover. However in a practical application this opportunity may not exist and there may be some ability for High-Rank Adaptation to help recover some of this lost audio. Models like Whisper [7] are already extremely resilient to noise, and completely missing words and phrases are difficult to recover. There does exist a small regime, in which only partially obscured words can disrupt the phoneme structure enough that fine tuning can potentially recover.

1 Introduction

My goal is to adapt Whisper to handle audio dropout to better handle audio information that has been corrupted. I was able to find a potential avenue of fine-tuning that would allow a large model like Whisper to be able to recover some degree of audio dropout. Specifically there exists a regime of audio dropout that is more than static noise but less than omitting entire phrases. When a single word or part of a word is omitted it can cause problems for a model like Whisper that are difficult to recover from. This I believe is because that degree of dropout is just enough to start to mess with the phoneme structure of the sentence which will then risk the Whisper model having difficulty interpreting. Through many dead ends trying to fine tune a model to address this, I was able to find a single glimmer of hope. This was by breaking Low-Rank Adaptation (LoRA) [3] into becoming high-rank adaptation instead of its' intended low-rank. My evaluations showed that there was a noticeable improvement over the base pre-trained Whisper-small model. In Section 5 I explain in detail what the metrics were, what they showed and specifically where I believe the real gain was expressed. I also

mention in Section 6 why I believe the results I got showed promise and what they actually mean for future work.

2 Background

Here I am trying to map an audio input to a text output. We are using a transformer [8] architecture so that we can retain the context that this type of model is good at. This has a self-attention mechanism that allows each position of the input to attend to all others in the sequence (which is how it gets the long range context). It also uses the mechanisms of keys, queries and values. The queries are essentially what the attention head is looking for. The key is the label of what it currently is representing. Finally the value is the assigned meaning once the model determines a match for the key and query. The model can then use multiple of these attention heads at the same time over the sequence. After all of the input the model will then autoregressively generate the output for what it thought the audio was. This gives us something that works in parallel, has long range context and can be scaled up with more layers, heads, and parameters. This is precisely what a model like Whisper needs to do speech recognition.

The model that I intended to build off was the Whisper-small model by OpenAI. This is a transformer based encoder/decoder for automatic speech recognition (ASR). This model was trained on a large dataset of diverse audio-text pairs. The model starts by processing the audio into a log-mel spectrogram. This is then encoded as a sequence into the model. From there the model uses byte-pair tokenization to produce the decoded output. The model is also very capable of generalization due to the broad distribution of training data.

I will define audio dropout as the zeroing out of segments of the audio signal before it is processed. This is to simulate missing or degraded audio infor-

mation that cannot be recovered. Within this audio are phonemes which are the smallest unit of sound that will distinguish words from one another. The audio dropouts I was concerned with were ones that would zero out enough of the audio to disrupt the phoneme structure in some way. This in turn should cause the pretrained Whisper model to fail in some cases.

2.1 Related Work

Whisper is relevant because it is the base model that I worked with extending. The quality of the audio input into the model is vital to getting good output. In contrast to pure text, there is much more that can go wrong with an audio signal that complicates the process of ASR. In general Whisper is claimed to be a robust model. It was trained on over 680K hours of weakly supervised audio-text pairs. Since the model was trained on such a diverse dataset they claim that it is robust to noise, accents, and domain shifts. The model is also intended to have zero-shot capability where it will perform well on tasks and languages not seen during training. All of this robustness however does not include structured audio dropout conditions. This is the specific failure mode that I was probing and attempting to make an extension of the base Whisper model to deal with.

Another work, Whisper-AT shows the limits of the robustness of Whisper, especially when noise overlaps with phonetic cues [1]. This is overlapping with what I was researching, since my controlled disruptions were targeting the phoneme structure of the sentence. However here they train a lightweight audio tagging model on top of Whisper. This is where my work differs, as their work does not specifically address the audio dropout issues.

Another paper "Speech recognition in adverse conditions: synthetic transformations and real environmental noise" [4] systematically evaluated modern ASR's under synthetic noise. The main findings concluded that larger models were more robust, in particular Whisper was very robust, but still vulnerable to extreme distortions. They also saw that synthetic distortions caused different failures than environmental noise. This paper was a good launch point for me to then take the next step into seeing if i could find some way to further mitigate these extreme distortions.

In order to try and adapt a Whisper model I was also going to need to make use of LoRA. This technique typically allows for low-rank adaptation

of larger models to efficiently fine tune them to another task. This is known as Parameter-Efficient-Fine-Tuning (PEFT) and is one of the most effective and widely adopted PEFT methods [2].

The work that I have done here differs because I am trying to specifically address the corruption of the phoneme structure of a sentence by building a fine tuned Whisper model. I am essentially building off the work done by Sergei Katkov, Antonio Liotta and Alessandro Vietti in their study and applying something similar. Here I am not doing a broad spectrum analysis but targetting one specific area highlighted in that paper and building a potential solution as well.

3 Data

For this experiment I used a subset of the Librispeech [6] dataset. This dataset is a collection of audio book like recordings that typically are around 5-15 seconds long. There are many different variants and i chose to focus on the train.clean.100 set which contains 28.5K clean audio transcripts. This alone can be used to test the performance of a model like Whisper, and all that was left for me to do was to add in the audio corruptions.

4 Model & Approach

The model that I built is a fine tuned version of Whisper-small that is used for a specific band of audio corruption. To do this i made use of LoRA used in an unintended way. This is supposed to be a parameter efficient fine tuning where you try to nudge the main model slightly in a direction that you want. This allows for new capabilities with a small amount of adjusted parameters. However, in order to get a good result I used LoRA in an unintended way by modifying the parameters to essentially make it High-Rank adaptation instead of Low-Rank. Even though this makes the process less efficient that it should be this still allows me to keep the function of the main model intact while still getting the results that I was looking for.

Figure 1 shows an overview of LoRA injection.

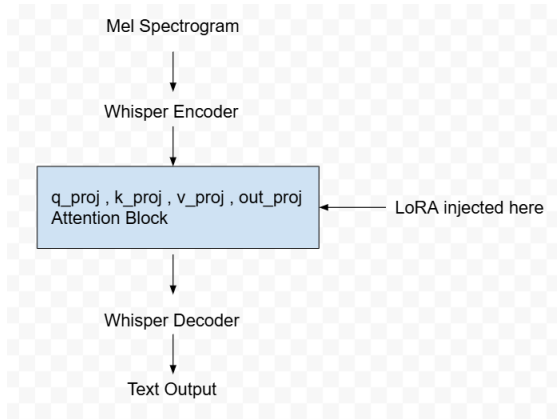


Figure 1: Schematic of where and what LoRA is injected into.

5 Evaluation

To evaluate this model I decided to take 20K of the librispeech train.clean.100 samples and then randomly selected 2,000 of them to corrupt and test. This gave me almost 10% of this dataset purely to test the fine tuned model on and compare to the Whisper-small pretrained. Here the Whisper-small pretrained model was acting as the baseline. I evaluated both models on controlled dropouts, where both models saw the same corruption on the same files and produced what they thought the result should be. This was a direct competition between the models. As a secondary evaluation I added a couple human trials with a completely fresh corruption of audio files, this time on only 10 samples of the 20K. This was mainly to get a temperature check on where a human interpretation might fit in here. I was not able to scale this test up enough to make any valid conclusions, but I think it gives some clear insight as well. All of these evaluations used only Character Error Rate (CER) and Word Error Rate (WER). These metrics are appropriate for ASR because it measures how closely the model output matches the true output.

5.1 Task and Procedure

To start testing my fine-tuned model, I first needed to instantiate the test data and then manipulate it. I used the train.clean.100 subset of the Librispeech dataset. Then for the next step of the pipeline I needed to shuffle a defined subset of this to collect and split that into 80% train, 10% test and 10% validate. Then from here I also define 20% of the training data to be my identity mapping of clean -> clean pairs. The remainder of the training data I would then corrupt using a TimeMask

from Audiomentations in which I would specify a millisecond dropout to perform and how many dropouts to perform per second of audio. The final audio dropout that was used for training was a single 300ms dropout for every 2.5s of audio applied randomly throughout the clip. This kept the ratio of zeroed out audio to clean audio the same, but also retained the random placement of the dropout.

The fine tuning used a high-rank LoRA configuration with $r = 512$ and $\text{lora_alpha} = 1024$, applied to the `q_proj`, `k_proj`, `v_proj`, and `out_proj` attention layers, with no LoRA dropout. The base Whisper-small weights remained frozen, and only the LoRA adapters were trained. Fine-tuning was performed for three epochs using a $1e-5$ learning rate, gradient accumulation, and mixed-precision (fp16) to keep training stable and efficient. Other training parameters followed standard Seq2Seq settings and did not really affect the adaptation. This configuration provided good updates while preserving the structure and behavior of the underlying model.

For the human evaluation, a CSV template was created on a random sample of the test split. The CSV file had the id of the audio file and the sentence that the person had to input their answer into.

For full evaluation in addition or supplemental to human evaluation, tests were ran on the full test set comparing the WER and CER. These were computed by running both the pretrained and fine-tuned Whisper-small models on the same corrupted audio samples, decoding each output, normalizing the text (lowercasing and removing punctuation), and then applying the standard jiwer implementations of word-error rate and character-error rate using the clean transcript as reference.

5.2 Metrics & Baselines

WER measures the proportion of word level substitutions, deletions, and insertions for the sentence being considered, while CER computes the same things at a character level. WER is the standard metric for evaluating overall transcript quality, however CER is more for granular distortions such as phoneme loss or character level deletions. Because dropout corruption removes short segments of audio and usually only disrupts parts of words, CER provides a more accurate indication of degradation, while WER is more useful at a broad level. As a guideline, error rates below 5% are considered excellent, values under 10% are considered acceptable, and rates above 10% generally indicate room

for significant improvement [5]. With regard to this, the baseline Whisper-small model is able to reliably achieve error rates under 5% on the clean samples of the dataset that I used. However it is important to note that the baseline performance was the model itself, not this number. This is because testing Whisper-small on the clean data verifies that it is a trustworthy model. We can then use that verified model as the best available when testing on the corrupt samples.

5.3 Results

Across the 2,000 corrupted evaluation samples, the fine-tuned model consistently outperformed the pre-trained Whisper-small baseline. WER decreased from 0.144 to 0.128 (about an 11% improvement), while CER dropped from 0.094 to 0.075 (about a 20% improvement).

Model	Accuracy
WER Pretrained	0.144481
WER Fine-Tuned	0.128611
CER Pretrained	0.093918
CER Fine-Tuned	0.075473

Table 1: WER and CER accuracy for pretrained and finetuned models.

This tells me that using a high-rank LoRA adaptation made meaningful improvements to the base model with respect to the ability to recover this type of audio drop out. Look through the results more thoroughly I can see that a large portion of this improvement seems to be the mitigation of catastrophic failures of the pretrained model. Figure 2 shows an example where the pretrained model ended early, where the fine-tuned model was able to keep interpreting the audio.

```

--- Example 458 ---
ID: 4499
Clean (Truth): joyce decided at once
               that she liked him oh did you know
               the family...
Pretrained Said: joyce decided at once
                 that she liked him
Finetuned Said:  joyce decided at once
                 that she liked him did you know her
                 family...
WER Pretrained: 0.72
WER Finetuned:  0.17
CER Pretrained: 0.75
CER Finetuned:  0.06

```

Figure 2: Example illustrating the fine-tuned model’s improved handling of dropout corruption.

6 Implications & Limitations

The implications for this are fairly narrow at the moment. The model fine tuning I have done only has an effect on a single specific use case. Any audio corruption that exists in short bursts and is frequent in the audio, appears to be more like static than omitted audio. This static is already well handled by the base pre-trained Whisper models and the fine-tuning I have done actually breaks the model in this respect. Especially considering that the WER and CER of the base model are negligibly affected when this occurs, my model is not to be used here. On the other side of the spectrum when you have entire phrases missing or even sometimes just entire words, both the fine-tuned model and the pre-trained base models start to make the same predictions, so here there is absolutely no change. With all this in mind there is just a single use case for my fine-tuned model, and that is when the audio corruption is more than just static, and it obscures parts of words to the point where the structure of the sentences are disrupted enough that the base model starts to suffer in accuracy.

7 Conclusion

Overall I was able to find a potential avenue of research where using High-Rank Adaptation may be able to help ASR models recover from audio dropout. This robustness is currently specifically targeted by a fine-tuned Whisper-small model. There was meaningful improvement to the WER and CER of the model after fine-tuning and typically produced the same results as the base model, with less of a chance of it catastrophically failing. For future work I would want to test this on a much larger scale. I was not able to get a well trained model on a larger dataset. This is the primary question I would want answered before any future work as it makes me suspicious of its functionality as a whole. The tests I ran suggest that it is promising but I would want to scale it up to be more certain it is true robustness. This matters because as others have shown, ASR models are vulnerable to extreme audio distortions, and there are plenty of ways for audio to become corrupted.

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